
Empowerment and team development

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Empowered teams and related concepts, such as autonomous task groups, self-managing groups or self-organization have received a lot of attention in theory and in practice (e.g., Dumaine, 1990; Manz, Keating and Donnellon, 1990; Mohrman, Cohen and Mohrman, 1995; De Sitter, Den Hertog and Dankbaar, 1997; Stewart and Manz, 1995). Self-managing teams are seen as a solution for organizational problems and are often introduced with the objective of simultaneously improving an organization's productivity, as well as the employees' quality of working life (Manz, 1992). The focus of this paper is on the development of self-organizing teams. First, we will briefly consider the concept of self-organization. We will do this by means of four principles of self-organization (Morgan, 1986):

- (1) requisite variety;
- (2) redundancy of functions;
- (3) minimal critical specification, and;
- (4) double loop learning.

Morgan may not be the first scholar who formulated these principles, but he brought them together within one framework, which we considered to be useful for the design of a plan to empower teams. In his work, Morgan did not explain how and in which order these principles have in practice to be introduced. Therefore, after outlining of Morgan's principles, we will relate these principles to team development. An implementation program will be described that was applied in a firm.

To help the members of the teams to get insight into the progress of the development of their team, we constructed a self-administered questionnaire. This instrument was developed during the first part of the implementation stretch and was used for the first time about two years after the first team started the empowerment plan. The results of this evaluation were discussed thoroughly with the team members and with management. Such feedback is part of most motivation and learning theories (e.g. Latham and Wexley, 1981;

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Locke and Latham, 1990) and therefore we considered this feedback to the teams as part of the change process itself. The feedback should help a team to get an impression of their own development and to obtain insight into the points on which further improvements may still be possible. It should help the team to express their priorities and to stimulate team discussion. On the basis of the “quantitative” evaluation data, the “qualitative” results of the team discussions, and our personal observations, we will consider the viability and applicability of the proposed intervention plan and try to find the factors that facilitate or hinder the empowerment process of teams.

Four principles of self-organization

The principle of “requisite variety” pertains to the rule that every team should have the capacity to transform the input of information, materials and parts into the desired output. This principle prescribes that variations from what is planned or expected should be controlled as closely as possible to the point of origin. In case of a lot of variety in transaction and transformation processes, the capacity level must be relatively high (e.g., Susman, 1976). Variety may refer to fluctuations in the demand for products or to changes in the mix of products. Variety may also refer to the extent to which products are customer-made and orders have unique characteristics, which will cause a lot of variety in the manufacturing or service process. In other words, this principle may provide an indication of the extent that control has to be decentralized and that the introduction of the other empowerment principles will be, from a management point of view, effective (see e.g. Niepce and Molleman, 1996).

The principle of the “redundancy of functions” states that within a task group, to enlarge the responsiveness of the group to fluctuations in the demand for its specific capabilities, there should be a redundancy of resources. With respect to human resources, redundancy can be created by developing the multifunctionality of workers. Through cross-training, workers become capable of performing a wide range of team tasks. This practice leads to job enlargement. It creates a buffer of capacity so that shifts in demand can be dealt with more effectively and, therefore, it increases flexibility. Moreover, it makes the group less vulnerable to fluctuations in the supply of resources caused by, for example, illness or turnover.

The principle of “minimal critical specification” has to do with the distribution of control within an organization and refers particularly to local autonomy and decentralized control. The principle implies that there should be as few restrictions as possible on the ways in which tasks are performed and just enough direction to ensure that the task is performed properly while allowing for the employee's personal contribution (cf. Cherns, 1987). In other words, upper management defines only the critical factors. As much autonomy as possible is given to employees to perform tasks according to their own knowledge and experience. Minimal critical specification refers to job enrichment in which workers gain control over activities that used to fall under the domain of management and, consequently a redistribution of power occurs

(Molleman and Van Knippenberg, 1995). The group takes over some managerial functions, e.g. the formulation of group goals, the internal distribution of tasks, and the way operations are performed. To realise these functions successfully, the self-reliance of the team, teamwork, and effective communication become important issues.

To understand the principle of “double loop learning”, we will first describe in a few words a condition without any learning and the process of single loop learning. Routine work refers to the cyclical process in which standardized “problems” (tasks) have to be solved, in which the best way to do the task is known and specified, and in which the required output has been precisely defined. If the output deviates from its standard, the actor is supposed to receive feedback indicating that he or she has deviated from the standardized work process and that the actor should correct his or her behaviour. In this situation there is no real learning process. In case of single loop learning, the output is still fixed, but in case of deviation the feedback may induce improved actions leading to the same results. Double loop learning implies the reconsidering of this single loop of “problem-action-output” learning. The problem will be reanalysed, the best work method is reappraised and even the required output may be reconsidered. This type of learning refers to redefining ineffective norms and setting new priorities (Argyris and Schön, 1996) and is especially relevant when there is a lot of variety in transaction and transformation processes that generates a lot of unique problems to be solved. Learning and, especially, double loop learning make demands on the learning and problem-solving capacity of the worker. To ensure that workers are able to solve successfully problems, it is necessary to develop problem-solving skills.

Team development

Theory building and empirical research on team development have their roots in social psychology and group dynamics (e.g. Cartwright and Zander, 1968). A flaw in this work is that it has mainly evolved in laboratory settings in which new groups were studied. We are interested in team development in field settings. Others have studied team development in practice, but have focused on emerging team development without explicit intervention (see for an overview McGrath and O'Conner, 1996), or on the relation between team characteristics and effectiveness (Campion, Medsker and Higgs, 1993). We, however, focus on team development with a specific objective: the development of self-organization in teams. Wellins, Byham and Wilson (1991) and Van Amelsvoort and Scholtens (1993) have dealt explicitly with the matter of team development in terms of empowerment and have distinguished several stages in this development. We want to complement and enrich their work by using Morgan's framework. We will use this framework, build on the work of Wellins *et al.*, and Van Amelsvoort and Scholtens, and supplement our study with our own experience in organizational development and empowerment (e.g., Niepce and Molleman, 1996; Born and Molleman, 1996). We will present a model of team development aimed at empowerment, in which we will distinguish four stages.

The word “stage” might suggest that we believe in a “linear-one-best-way” form of team development, on which the beginning and the end of each stage are clearly marked. However, we believe that there are many ways to empower teams, that their stages may be mixed, and that their sequences may change. Moreover, a stage approach suggests that team development is an ending story, while in a dynamic environment its better to regard it in a perspective of “continuous improvement”. Nevertheless, to empower teams, one cannot focus on all aspects at once. Therefore, we have specified four stages, in which we have ordered the issues with which a team has to deal from the least to the most complex (see Figure 1). Wellins *et al.*, (1991, p. 26-28) also speak of arranging tasks in order of their complexity and of lower- and higher-order responsibilities, of which any actual sequence is organization specific. With respect to complexity, we will distinguish three interrelated dimensions, i.e., we will shift our attention from routine tasks to non-routine tasks, from individual to group, and from inwards oriented to outwards directed.

In the first stage, the team concentrates on the “redundancy of functions”. Qualification matrices are made and a cross-training program is generated to develop multifunctionality and to support job enlargement in the light of the variety in the demand for production capabilities and the supply of work skills (“requisite variety”). In the next stage, the focus is on the principle of minimal critical specification, and thus on the redesign of the control structure. As far as possible, control tasks are delegated to the team and the focus is on job enrichment. In these first two stages, the focus is on the individual and there is a shift from routine to non-routine activities. In the third stage the emphasis is on the self-reliance of the team, on teamwork and on team-building communication and decision-making. During daily work the team has to become independent of its supervisor. As a result, the focus is on teamwork and autonomy. In the last stage, in line with the principle of double loop learning, the capacity to solve the most non-routine problems is the main concern. An important area in which this kind of problems often arises is the management of the environment by the team. Therefore, we supplemented a theme, which is not explicitly covered by Morgan's principles, but which we suppose to be an important addition: the management of team boundaries. This refers to the handling of relations with other groups or individuals who provide inputs or receive outputs from the team (Katz and Kahn, 1978).

The activities during the first three stages are mainly directed inwards, leading to a higher level of empowerment of the team. Group boundaries are emphasised and this will motivate group members to strengthen group identity (e.g., Tajfel, 1981). Sharp group boundaries often strengthen the group norms and reduce the tolerance for deviant attitudes and behaviour among group members. This may limit the innovative and creative capacity of the group, and may reduce its responsiveness to suppliers and customers, which is an important reason for empowerment. An autonomous working team might become too inwardly oriented, a tendency that may constrain organizational learning and overall system improvement (e.g. Adler and Cole, 1993). These

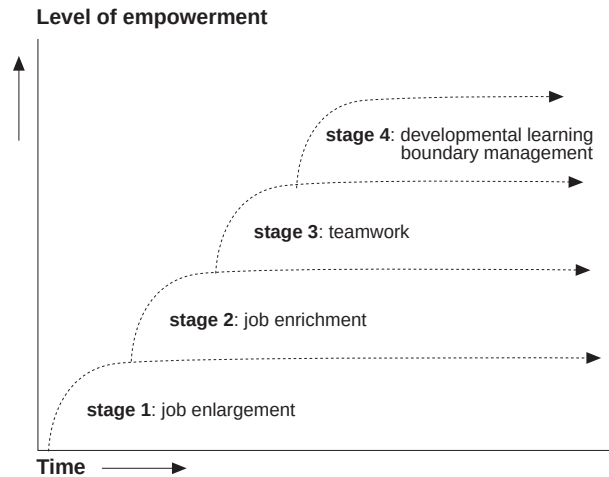


Figure 1.
A model of team
development

arguments suggest that team boundaries should neither be too rigid nor too loose. To find the right balance, the notion of boundary management was included in the fourth stage, with the purpose of creating teams that are not only autonomous, but also responsive, pro-active and innovative.

In general, the model strives to empower teams through job enlargement (stage 1), job enrichment (stage 2), teamwork (stage 3), “self improvement” and boundary management (stage 4). The focus is successively on (1) primary processes, (2) the control structure, (3) team building and internal communication, and (4) developmental learning and the management of team environment. The objective of this study is to gain insight into the viability and the applicability of our model and to find relevant factors that facilitate or hinder the empowerment of teams in a field setting. In the next section we will first give a depiction of the firm and its teams and then we will briefly discuss the instrument we have developed.

Method

In this section, we will first give a general characterization of the firm and then present some details of the teams, which developments we evaluated. Next, we will describe the parts of the questionnaire and highlight the procedure we applied.

The firm

The firm is a leading company in the production of high quality industrial glass, which is used in a wide range of products, for example, in halogen lamps, in sport-field illumination, in solarium, in horticulture, and in car lamps. At present the production range spans some 2000 different end products fabricated from around 30 different types of glass. The wide variety of products and materials, accompanied by an efficient, high-quality industrial production,

requires a high level of skills in the workforce and state-of-the-art equipment. The plant is part of a large multinational industrial group. The staffing-level of the plant is about 380. The production process can be divided in two main steps. During the first step sand is smelted and moulded into long glass pipes. This part of the plant is called the “pulling” unit. The main characteristics of the production process are like those in the process industry. In the second step the glass pipes are washed, sawed, sorted and controlled. This department is named the “finishing” department and its manufacturing may be typified as batch oriented.

At the beginning of 1994, it was first proposed by management that empowered teams should be developed. The need to create such teams may be regarded primarily as a response to environmental changes, which demanded greater flexibility. Customer demands became increasingly more diverse and emphasised smaller batches, higher quality, shorter delivery times, and lower prices.

This study focuses on five teams. Two of them are part of the “pulling unit” and the other three belong to the “finishing” department. These teams range in size from 4 to 15 members. To each team a mentor was added, who had to support the team to be empowered. The five groups did not start at the same time. The first team started about 24 months before the questionnaire was filled out for the first time, another team about 20 months, two more teams about one year, and the final team about 3 months. The starting points were varied to prevent the organization from being troubled by too much turbulence. The other parts of the plant will enter the empowerment program in the near future. At the time when the first team started, there was no elaborated plan, and during the first period there were no explicitly defined goals and implementation strategies. Such a method can best be typified as “learning by trial and error” or “learning by doing” and is compatible with an “organizational learning” strategy and a bottom-up approach that is one of the goals of empowerment. In time, concrete ideas were formulated and experiences were also communicated to all the locations that started the program at a later point in time. A top-down expert approach was not used, and the model described briefly in the introductory part of this paper was actually developed during implementation.

The questionnaire

The design of the questionnaire was based on our critical reading of the literature (Wellins *et al.*, 1991; Campion *et al.*, 1993) and on suggestions from the team members. The questionnaire consists of 5-point Likert-type items. To measure the level of job enlargement, we compiled a list of 20 items, e.g., “Team members are able to take over each other's jobs” and “Tasks are rotated among team members”. For “job enrichment” we formulated 24 items, e.g., “The team is responsible for quality control” and “Detail planning is done by the team”. To appraise the developments in the third stage, 15 items were formulated, e.g., “Team members motivate each other to do a good job” and “The team solves

daily work problems without interference from the supervisor". To find an indication for the progress in the last stage we formulated 19 items, e.g., "Team members initiate improvements" and "The team itself deals with customer complaints". The Cronbach's alpha's for the four scales range from 0.82 to 0.88.

About two years after the first team started the implementation, data were collected for the first time. The members of the five teams filled out the questionnaire, and the results were discussed with the team members during one of their regular meetings. The questionnaire was designed to be used as a self-diagnostic instrument, and therefore the format of the feedback returned to the teams was specified according to the preferences of the teams. Teams preferred feedback in percentages above means; the sum of the percentages of team members that "agreed" and "agreed strongly" was reported. Also standard deviations were reported, and we volunteered to discuss scales and items with low percentages or high standard deviations in team meetings. In four teams the questionnaires were filled out individually during team meetings, however, the members of fifth team preferred to do this outer their meeting. There were 52 workers in the five teams, and 46 (88 per cent) completed the questionnaire. Three workers were ill, and three workers did not respond for another reason. These last three workers were all members of the fifth team, and thus this non-response may be related to the procedure that we applied. This team did not fill out the questionnaire during a team meeting and hence, there was less control over participation, which may be a possible reason for this non-response. In the next section we will present the results from the questionnaire. We have discussed these outcomes with the teams, and we will use their reactions to interpret the results in a more qualitative way.

Results

We will first discuss some overall findings and then we will focus in a more detailed way on the differences between teams and the developments within each team. In the next section we will formulate general, as well as specific conclusions and deal with some practical implications of this study.

In Figure 2, the scores of the teams on each of the four scales are shown for the five teams separately. For each team, the developments in each stage are presented. We may infer from this figure that, with one exception (team 5), the progress corresponds with the sequence in which developments were introduced. The job enlargement (stage 1) is the most developed feature in all the teams, and team enrichment is the second most developed feature (stage 2). Overall, the stage 3 characteristics are less and the stage 4 qualities are the least developed. We see these results as support for our stage-like approach to the empowerment of teams. However, we have to ascertain that even stage 1 has not been developed to the maximum level and that teams seem to be working on elements from other stages at the same time.

Based on our discussions with team members, we found three issues which might decrease the validity of our stage model. First, the sequence in which issues are emphasized is on the one hand felt to be helpful in structuring the

implementation process: from routine to non-routine, from individual to group, and from inside to outwards directed. On the other hand, it is perceived as somewhat artificial; the notions involved in job enlargement, job enrichment and group dynamics are not completely independent and job enlargement, for example, cannot be introduced completely without dealing with elements of control or group dynamics. A closer look at the responses presented in Figure 2 shows that job enlargement and job enrichment go more or less hand in hand. Second, as we have stated in the introduction, in the light of “continuous improvement” a stage may never end, and therefore a score of 100 per cent will never be reached. Third, in the teams we studied the variety in transactions and transformations is limited and therefore, with the principle of “requisite variety” in mind, the need to empower teams is confined. The requisite level of self-organization is related to environmental and production characteristics, and, from a management point of view, it will not be necessary to obtain the maximum scores on each of the four scales.

Based on the results shown in Figure 2 and the reactions of the respondents to this feedback, we conclude that the focus of the teams at the time of measurement was in general on the distribution of control tasks (stage 2) and on group-dynamic issues (stage 3). The developments with respect to problem solving capacity, and boundary management in particular, are, except for team 3, small or rudimentary. Indeed, most workers perceived the enlargement of problem solving capacity as the most complex problem and boundary management matters as the most delicate one. Bargaining and making agreements with suppliers and buyers inside and outside the plant, not only requires high-level skills but also the explicit support of upper management.

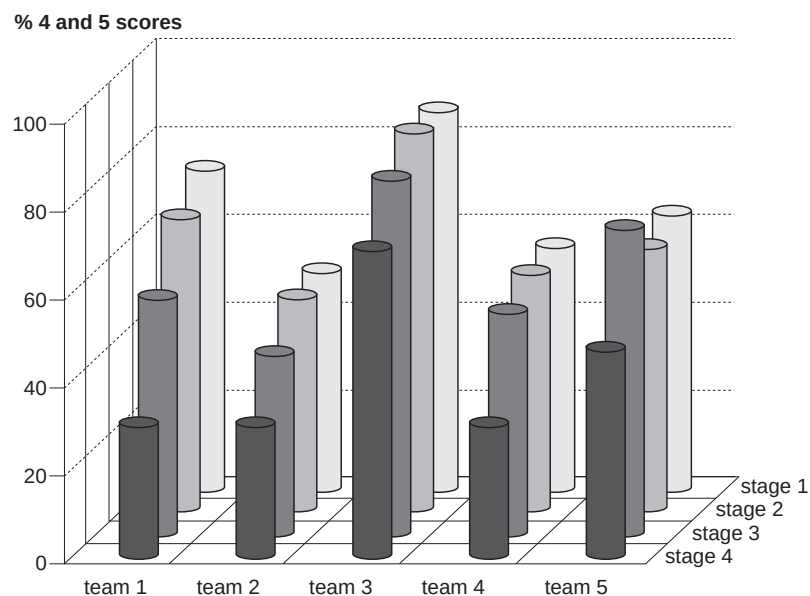


Figure 2.
Team scores and
developments at stages
1-4

Boundary management exposes the successes and the failures of a team to outsiders, which is experienced as threatening by part of the workers.

Looking at the individual teams, we may infer from Figure 2 that team 3 is the most developed team (with respect to all four phases). This is the smallest team with only 4 operators functioning within the “finishing” department. The team started the empowerment program about 20 months ago. This team is responsible for the more complex processes and uses the most advanced equipment. The mean batch-size is smaller than in the other teams and many special products are processed in this group. There seems to be a lot of variety, and thus much to empower. We think that the success of this team can also be attributed partly to the way in which the mentor supports the team and his style of communicating. He encourages all the team members to participate actively in decision making, is a good listener and takes care that all team members make more or less equal contribution to decision making. In dealing with the “inside” issues, he is a person who asks more questions than he gives answers. With respect to environmental issues, he informs the team of important developments in and outside the plant and makes the team responsive for external changes. The level of multifunctionality is high, and all the qualifications of members are actually used by systematically rotating jobs.

Team 1 and team 5 are quite a distance behind team 3 in their development as empowered groups. Team 1, which started first, has reached a rather high level of job enlargement and job enrichment. This team consists of 9 workers and is part of the “finishing” department. In the areas of team work and group dynamics, we observed four possible obstacles to further development. The team members differ in their attitudes toward empowerment. In particular, one member is a strong supporter of empowerment and actively campaigns for it. For example, he is often asked by non-team members to present his experiences of empowerment to outsiders. Second, the mentor of this team is a line manager, who is strongly involved in the empowerment program, but his enthusiasm and directive management style seems to hinder the initiatives of the (other) team members. Third, a couple of months ago this team was down-sized from 12 to 9 workers on the basis of a decision sent down from upper management. This “top-down” display of power may partly explain the closed nature of this team and their negative attitude towards the environment. Such characteristics are reflected in the low scores on boundary management, which has hardly begun to develop in team 1 (see Figure 2). Fourth, there are in fact two sub-groups in this team. One is operating the sawing machines and the other sorts and controls the products. There is job-rotation within, but not between, these sub-groups.

Team 5 has 15 persons and is part of the “pulling unit”. The mean age of this team is higher than that of the other teams. Most members have worked for 15 years or more in this plant and even in the same group. This team started the empowerment program about 3 month ago. On the one hand this team has rarely any problems, but on the other hand its stability makes it rather insensitive to change. Most team members indeed declare that since the start of

the empowerment project not much has been changed. We think that the scores of this team, presented in Figure 2, should mainly be attributed to its accumulated work experience. Group dynamics are rather well developed, and the experience of the members makes the team relatively strong in boundary management.

Team 4 and especially team 2 are less well developed than the other three groups. Both teams started their empowerment project one year ago. Team 4 is part of the “pulling” unit and consists of 15 members. The team members state that they are motivated to empower themselves, but that their workload is so high that they find hardly any time to think about “what” and “how” to change. Moreover, they declare that management is often interfering in daily work by, for example, asking for rush orders, which limits the potential for decentralized control. It makes the team members somewhat suspicious about the “real” attitude of management to empowerment, and it makes them feel incompetent in managing their boundaries.

Team 2 has 9 members and belongs to the “finishing” department. This team is the result of a fusion of two teams that occurred about a year ago. This fusion was accompanied by a rather radical down-sizing. Moreover, over the last year, the composition of this team has changed several times as a result of such things as employee turnover. Up to a couple of month ago, this team was supported by a mentor who was overly focused on team performance and had less affinity with the group dynamic side of team empowerment. Recently, they also have been supported by a member from the personnel department, who is more aware of the human aspects. Another problem of this team is the exceptionally negative attitude towards empowerment that one of its members has, and his presence seems to have a negative impact on the other team members.

Conclusions and discussion

We have formulated a model of team empowerment with the help of the principles outlined by Morgan (1986) and the work of Wellins *et al.*, (1991) on the empowerment of teams. We have ranked the main themes in an ascending order of complexity. The easiest issues are dealt with first and the most complex last. The focus is respectively on job enlargement (“redundancy of functions”), job enrichment (“minimal critical specification”), team autonomy and group dynamics, problem solving capacity (“double loop learning”) and boundary management. The teams indicate that a step-by-step introduction of empowerment facilitates team development, which is supported by the quantitative data. However, we have seen that while a specific step is still unfinished, teams are already working on issues from the next stage. Items from different steps are, to some extent, interdependent and can not be completely separated. Moreover, empowerment is an ongoing process, and therefore, any stage of development will never be totally completed. We conclude that Morgan's principles, as well as the sequence in which we have put

them, are helpful in developing empowered teams. However, they should not be used as an end in itself.

Besides these more general conclusions about our model, the responses of the participants have raised several issues which may facilitate or hinder the development of empowerment (Figure 3). The extent to which the teams are empowered and the time elapsed since they started the empowering program are only moderately related. In fact the correlation between the number of months engaged in the empowerment program and the total score over the four stages is $r=0.37$. This moderate association may partly be attributed to the implementation strategy. The first teams learned by doing and the teams which started later also learned from the experiences of those who started first. However, we think unique factors and conditions also contribute to this moderate relationship. We will discuss these factors next.

The role of management

The first factor that seems to be of importance for team empowerment of teams is the role of management. Our study suggests that to develop empowered groups successfully, management should promote the empowerment programs by words and, especially, by actions. Top-down interventions by top or upper management may dramatically undermine empowerment. It will especially inhibit the decentralization of control tasks and the development of boundary management. It may even create distrust or even hostility towards management or towards the outer world in general. However, as our study has also shown, the empowerment of teams does not take place in a vacuum and top-down interventions may sometimes be inevitable. To stay competitive down-sizing is, for example, sometimes unavoidable, and it seems to be unrealistic to ask teams to down-size themselves or each other. Regardless of such extreme situations, the experiences of team 4 show that the development of empowerment “takes time”, and if management gives a team a very high workload, it will become frustrated. Our findings also show that the behavioral style of middle and lower management is important. These managers should display a good mix of team and task oriented behaviour. Their style should, like the mentor in team 3, be participative and facilitative in nature. In fact, the style itself has to be compatible with the outcomes at which it aims (cf. Emery, 1993). Nevertheless,

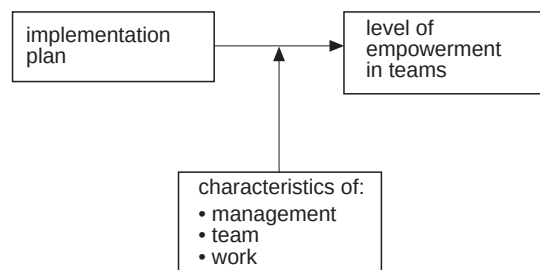


Figure 3.
Factors influencing the
development of
empowerment

we realize that "helping to become autonomous" sounds somewhat paradoxical and that such help might easily create some dependency.

Group dynamics

With respect to the developments within the team, group dynamics is another important concern. Our observations indicate that the homogeneity of the team on the issues that are related to the development of empowerment may facilitate empowerment in several ways. There should be homogenous attitudes about the development of empowerment. As we saw in team 1, even excessively positive attitudes may, if they deviate from the group, have a negative impact on team development. Next, the existence of subgroups seems to be undesirable. The subgroups in team 1 desire to strengthen their own identity and to distinguish, in a positive sense, their own subgroup from the others. This finding accords with other studies on intergroup relations (see, for example, Tajfel, 1981). The subgroups in team 1 each have their own task, and they perceive that their own task has more significance and greater status than that of the other subgroup. They, therefore, dislike job-rotation between the tasks of each subgroup (see also Cordery, Sevaston, Mueller and Parker, 1993). The developments in team 5, however, suggest that teams can also be excessively homogeneous. Therefore, we conclude that group cohesion contributes to the development of empowerment in teams, but that it should not be too strong. A certain (low) level of attrition or a small amount of rotation of workers among teams may be desirable in order to keep teams open to change, to stimulate (organizational) learning, and to facilitate the dissemination of experiences.

Work characteristics

Finally, we want to emphasize that the empowerment of teams is, as Ford and Fottler (1995) have stated, a "matter of degree". The principle of "requisite variety" supposes that when there are many fluctuations in the demand for products or changes in the mix of products, boundary management and multifunctionality become – referring to the characteristics of the manufacturing processes in team 3 – important issues. If variety in the manufacturing process itself is high, decentralized control and the problem solving capacity of teams become important matters. From a management point of view, empowerment must be made contingent on the variety of a company's transactions and transformations, which determine themselves the nature of the work.

Limitations of the study

The foregoing conclusions are only tentative and should, for several reasons, be cautiously considered. First, the number of observations is too small to evaluate the psychometric characteristics of our instrument in detail, and the reliability coefficients presented will be rather unstable. Second, it should be clear that the internal validity of our study is too weak to make valid causal inferences. A valid study of (team) development should somehow be longitudinal in nature,

and at least our study should have had some base-line or "pretest" data to support our inferences about the development of empowered groups. Third, the external validity of our study is limited. We only studied five teams within one firm, and some items in our questionnaire were operationalized in a very context specific way, so that they, for example, referred to certain firm-specific control tasks. We think that the need and possibilities for empowerment are determined by the variety in an organization's transactions and transformations (see Niepce and Molleman, 1998). Therefore, we suppose that our findings may be limited to organizations with more or less the same environmental and production characteristics. Because of these flaws, our conclusions should be regarded as suggestions for further research on the subject of team empowerment.

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